

Discrete-Time Design of Adaptive Current Controller for Interior Permanent Magnet Synchronous Motors (IPMSM) with High Magnetic Saturation

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Abstract- IPMSM for automotive traction applications exhibits a rather high magnetic saturation as a result of the maximization of power and torque densities. Additionally due to the high speed range high electrical frequencies in proportion to the converter switching frequencies occur. That results in a small number of samplings per electrical fundamental. Both effects have to be taken into account to design a suitable current controller. In this paper a discrete-time IPMSM model considering parameter variations due to magnetic saturation as well as the effect of a small number of sampling instants per electrical rotation is derived. Then the design of an adaptive current controller is presented and evaluated by experimental results.

INTRODUCTION

Interior permanent magnet synchronous motors (IPMSM) provide high power and torque densities as well as a very high efficiency. These characteristics are crucial for applications with limited installation space e.g. in the automotive industry. Thus the IPMSM is often used as a traction drive in today's modern electric vehicles (EV) and hybrid electric vehicles (HEV). Apart from a nonlinear reluctance torque due to asymmetrical reluctances and a wide constant-power operation range, a high amount of magnetic saturation is accepted in automotive applications. Since typically high numbers of pole pairs are preferred, the electrical frequency reaches values of up to 1 kHz at a typical converter switching frequency of about 10 kHz, which results in a small number of samplings per electrical fundamental at high speeds. To design a drive controller that provides stability as well as good dynamics and disturbance rejection in the entire operation range, both effects have to be taken into account. However the main objective of the drive controller is to provide the requested torque with adequate accuracy and especially high efficiency.

A suitable control scheme is presented in Fig. 1 [1]. It is realized as a cascaded control structure with a field-oriented current controller as inner control loop. In addition, some auxiliary elements are required to ensure stability and high efficiency. Appropriate operation points must be chosen to provide the requested torque that follows from the drivers demand and superimposed controls with high efficiency. Due to magnetic saturation the torque generation is strongly

nonlinear and can hardly be analytically described. Thus, the torque control is typically realized based on lookup tables [2], [3], [4], [5]. In the constant-power operation range, the maximal torque is limited by the available voltage. Nevertheless, a suitable voltage reserve is required to ensure the stability of the inner current control loop. If the voltage reserve is maintained in terms of an open-loop control, a safety margin has to be provided due to model uncertainties. Instead, a modulation controller (voltage controller) should rather be used to utilize the DC-bus voltage on the one hand and to retain a suitable voltage reserve on the other hand [1].

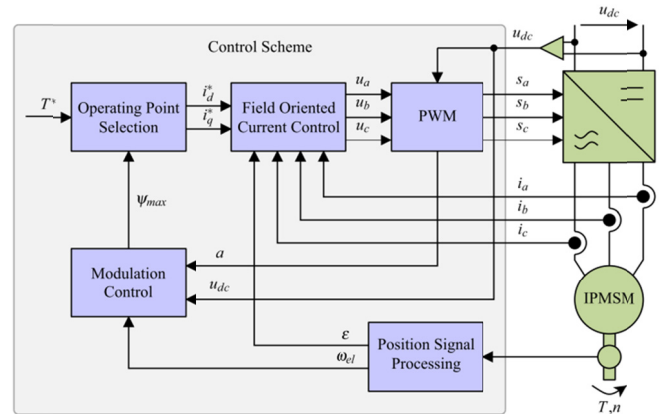


Fig. 1: Control Scheme Overview

In this paper the focus is on the design of the inner current control loop. A discrete-time IPMSM model is presented taking into account the effect of a small number of samplings per electrical fundamental as well as magnetic saturation. Based on that model, a field-oriented current control is designed and the impact of parameter variations due to magnetic saturation is shown. Then the adaption of the controller parameters is investigated considering small-signal and large-signal conditions. The control is realized and evaluated by experimental results.

DISCRETE-TIME IPMSM MODEL

The discrete-time IPMSM model is derived from the continuous flux differential equation in the stator reference frame (1).

$$\frac{d\psi_{\alpha\beta}(t)}{dt} = \underline{u}_{\alpha\beta}(t) - R\underline{i}_{\alpha\beta}(t) \quad (1)$$

It is assumed that the control is synchronized with the pulse width modulation, which is known as regular sampling. Thus the current components $\underline{i}_{dq}[k]$, the flux components $\underline{\psi}_{dq}[k]$ and the rotor position $\varepsilon[k]$ are the momentary values at the time instants $t = kT_a$ while the voltage components $\underline{u}_{dq}[k]$ are understood as averaged values of the time interval $t = [kT_a \dots (k+1)T_a]$. Since the flux derivative is approximated by a difference of sampled flux values that correspond to different rotor position values, a suitable discrete-time model is obtained by discretization of (1) in the stator reference frame using the 1st order Euler method and transformation of the resulting discrete-time model into the rotor reference frame [1], [6]. As presented in [1], the resulting difference equation in the rotating reference frame is given by (2). The left hand side of (2) describes the flux difference in one sampling period and the right hand side contains the applied voltage rotated by the angle $-\Delta\varepsilon[k] = -(\varepsilon[k+1] - \varepsilon[k])$, the resistive voltage drops and the induced voltage (EMF).

$$\begin{aligned} \frac{1}{T_a}(\underline{\psi}_{dq}[k+1] - \underline{\psi}_{dq}[k]) &= \underline{Q}(-\Delta\varepsilon[k])\underline{u}_{dq}[k] \\ -R\underline{i}_{dq}[k] - \frac{1}{T_a}(\underline{E} - \underline{Q}(-\Delta\varepsilon[k]))\underline{\psi}_{dq}[k] \end{aligned} \quad (2)$$

Due to saturation and cross-saturation effects there is a strongly nonlinear relationship between the flux linkage and the current. Thus the induced voltage is directly calculated from the actual flux values that are determined from the measured currents using lookup tables according to (3).

$$\underline{\psi}_{dq}[k] = f(\underline{i}_{dq}[k]) = \begin{bmatrix} \psi_d(i_d[k], i_q[k]) \\ \psi_q(i_d[k], i_q[k]) \end{bmatrix} \quad (3)$$

To describe the current dynamics, the flux difference $\Delta\underline{\psi}_{dq}[k] = \underline{\psi}_{dq}[k+1] - \underline{\psi}_{dq}[k]$ has to be expressed as a function of the current difference $\Delta\underline{i}_{dq}[k] = \underline{i}_{dq}[k+1] - \underline{i}_{dq}[k]$. Therefore, the inductances can be calculated from the flux gradient between the operation points in the i_d, i_q -plane at two successive sampling instants according to (4) and (5).

$$\Delta\underline{\psi}_{dq}[k] = \begin{bmatrix} L_{dd,\Delta}[k] & L_{dq,\Delta}[k] \\ L_{qd,\Delta}[k] & L_{qq,\Delta}[k] \end{bmatrix} \Delta\underline{i}_{dq}[k] \quad (4)$$

$$L_{xy,\Delta}[k] = \frac{\psi_x[k+1] - \psi_x[k]}{i_y[k+1] - i_y[k]} \quad (5)$$

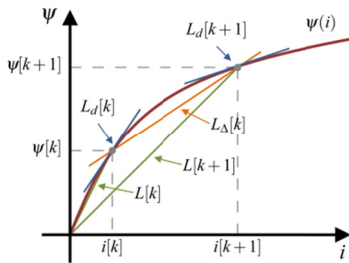


Fig. 2: Magnetization Curve and Definition of Inductances

The interpretation of these inductances is illustrated in Fig. 2. For every operating point on the magnetization curve an

absolute inductance $L[k]$ as well as a differential inductance $L_d[k]$ can be defined. In addition a straight line through two successive operation points can be interpreted as an equivalent inductance $L_\Delta[k]$, which converges to the differential inductances in steady state operation. For the IPMSM with a two-dimensional magnetization map, the differential inductances are given by the Jacobian in (6), which is the derivative of the flux with respect to the current. The diagonal entries of $\underline{L}_{dq}$ are relevant for the current dynamics and the off-diagonal entries describe the cross-couplings due to cross-saturation effects.

$$\underline{L}_{dq}(\underline{i}_{dq}[k]) = \begin{bmatrix} L_{dd}[k] & L_{dq}[k] \\ L_{qd}[k] & L_{qq}[k] \end{bmatrix} = \begin{bmatrix} \frac{\partial \psi_d}{\partial i_d} & \frac{\partial \psi_d}{\partial i_q} \\ \frac{\partial \psi_q}{\partial i_d} & \frac{\partial \psi_q}{\partial i_q} \end{bmatrix} \quad (6)$$

Since the cross-coupling terms of $\underline{L}_{dq}$ are typically smaller than the diagonal elements by a factor of about 10, they are neglected and considered as disturbances in the current control loop. The resulting discrete-time IPMSM model shown in Fig. 3 consists of two PT1-blocks that are coupled by the induced voltage and the rotation of the input voltage. The induced voltage is accurately described using the flux vector $\underline{\psi}_{dq}[k]$ including saturation effects. Due to the rotation of the input voltage by the angle $-\Delta\varepsilon[k]$ and the discrete-time description of the induced voltage the discrete-time model is quite accurate even if the rotation angle of one time step is considerably large. The dynamics of the PT1-elements are approximated using the diagonal elements of the differential inductance matrix (6).

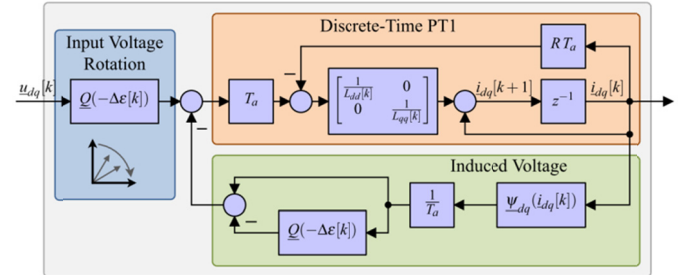


Fig. 3: Discrete-Time IPMSM Model

CURRENT CONTROLLER DESIGN

Based on the presented discrete-time IPMSM model a suitable control scheme is presented in Fig. 4. To provide a proper decoupling of the current components i_d and i_q the induced voltages are added to the controller output and the resulting sum is rotated by the angle $\Delta\varepsilon$. The induced voltages are calculated based on the flux vector $\underline{\psi}_{dq}(\underline{i}_{dq}[k])$, which is determined using lookup tables gained by measurements including saturation effects. Due to the controller dead time the calculated induced voltages are delayed. Since the voltages applied to the motor in the current sampling period are known, a flux and current predictor can be used to eliminate this delay. The rotation angle $\Delta\varepsilon$ only slightly changes during one sampling period.

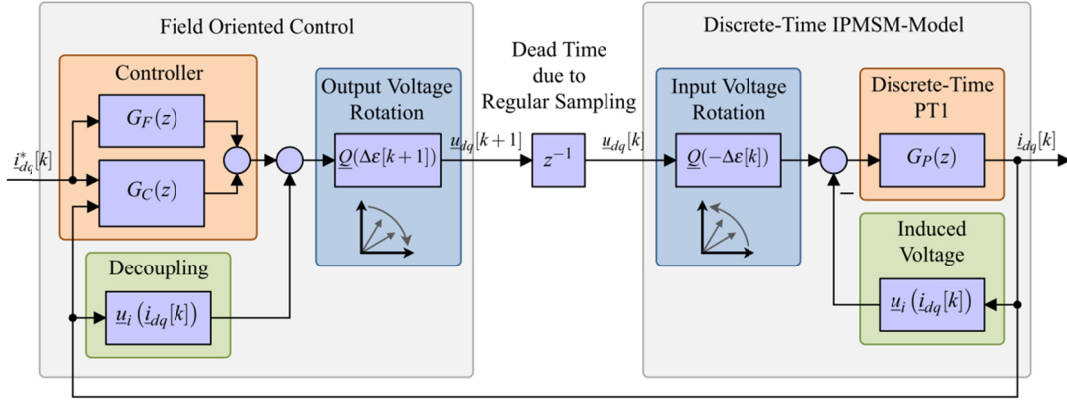


Fig. 4: Field Oriented Current Control with Decoupling

Two separate current controllers can be designed for the current components i_d and i_q since the resulting plant consists of two independent discrete-time PT1-elements with transfer functions given by (7) with the controller sampling period T_s . Note that the plant time constant T_1 varies due to saturation effects and thus the function $G_P(z)$ is regarded as the small-signal plant model.

$$G_P(z) = \frac{K_{PP}}{\frac{T_1}{T_s}(z-1) + 1}; K_{PP} = \frac{1}{R}; T_1 = \frac{L}{R} \quad (7)$$

The two current controllers consist of a PI-controller with the transfer function given in (8) and a reference feed forward path with a proportional gain $G_F(z) = K_F$ each. Note that the controller dead time resulting from regular sampling has to be considered to calculate appropriate controller parameters.

$$G_C(z) = K_P + K_I \frac{1}{z-1}; K_I = K_P \frac{T_s}{T_I} \quad (8)$$

The resulting closed-loop transfer functions are given by (9) and can also be represented in terms of poles ($\alpha_1, \alpha_2, \alpha_3$) and zeros (β) according to (10). The controller parameters can be calculated using the pole placement method.

$$G_R(z) = \frac{G_C(z)G_P(z)z^{-1} + G_F(z)G_P(z)z^{-1}}{1 + G_C(z)G_P(z)z^{-1}} \quad (9)$$

$$G_R(z) = K_R \frac{(z-\beta)}{(z-\alpha_1)(z-\alpha_2)(z-\alpha_3)} \quad (10)$$

The poles of the closed-loop transfer function are the roots of a cubic function and thus the following solutions may exist: one real and two complex conjugate poles, three real poles with a multiple pole, three identical real poles or three distinct real poles. From control theory view the denominator of (9) consists of a PT1-element with a real pole and a PT2-element with real or complex conjugate poles in series. With the method of equating the coefficients the relationship between the coefficients of (10) and the controller parameters in (9) can be determined. Since the closed-loop transfer function has three poles but only two variable parameters, which are the PI-controller gains K_P and K_I , the poles can only be set according to the boundary condition in (11).

$$\alpha_1 + \alpha_2 + \alpha_3 = 2 - \frac{T_s}{T_1} \quad (11)$$

The resulting PI-controller parameters are given in (12) and (13). After the PI-controller parameters are determined the zero β can be placed using the proportional gain of the reference feed forward path according to (14).

$$K_P = R \frac{T_1}{T_s} (\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_3 \alpha_1 - 1) + R \quad (12)$$

$$K_I = R \frac{T_1}{T_s} (\alpha_1 \alpha_2 + \alpha_2 \alpha_3 + \alpha_3 \alpha_1 - \alpha_1 \alpha_2 \alpha_3 - 1) + R \quad (13)$$

$$K_F = \frac{K_P(\beta - 1) + K_I}{(\beta - 1)} \quad (14)$$

To avoid overshoot three real poles are chosen. Due to the restrictions resulting from the boundary condition (11) a good closed-loop dynamic is gained when the three poles and the zero are chosen identical according to (15). If smaller values than the value resulting from (15) are assigned to two of the poles the value of the third pole increases according to (11) and can even become unstable. Although one stable pole with a high value resulting in slow dynamics can be compensated using the zero in the closed loop transfer function it still would cause a bad disturbance rejection.

$$\beta = \alpha_1 = \alpha_2 = \alpha_3 = \frac{1}{3} \left(2 - \frac{T_s}{T_1} \right) \quad (15)$$

ADAPTATION OF CONTROLLER PARAMETERS

In the control scheme presented in the last paragraph saturation effects are taken into account in the calculation of the induced voltages for the decoupling of the current components. In addition the plant time constant T_1 may vary considerably due to saturation effects. If fixed controller gains are calculated based on a constant plant time constant the resulting pole locations are varying if the plant time constant changes due to saturation effects. Supposing the controller is still stable and the resulting pole locations are acceptable, a controller that is robust to parameter variations due to saturation effects is gained. Otherwise if the control becomes unstable or poles with strongly oscillating behavior occur, the controller design has to be revised. In this case the controller parameters can be adapted to the varying plant time constant, which is to be investigated in the following.

To exactly adapt the controller parameters to the plant time constant the actual inductances have to be known. Note that the inductances $L_{da,\Delta}[k]$ and $L_{dq,\Delta}[k]$ in (4) and (5) depend on the currents at the sampling instants k and $k+1$. Additionally due to the controller dead time the currents at the time instants $k+1$ and $k+2$ have to be known for an exact consideration of the varying plant time constant. Since the voltage that is applied to the motor in the current sampling period was calculated in the last sampling period and therefore is known, the current and flux values at the time instant $k+1$ can be predicted using the presented IPMSM model. However the current and flux values at the time instant $k+2$ are not known and thus the plant time constant is also not known exactly in transient operation. Therefore, a suboptimal adaption of the controller parameters is presented and the controller parameters are adapted in a way that the controller is robust to unknown inductance variations. While the resulting poles would be exactly kept at the specified locations in case of an exact gain adaption, some pole location variation is accepted in the proposed approach but the poles are kept within a defined area around the specified location.

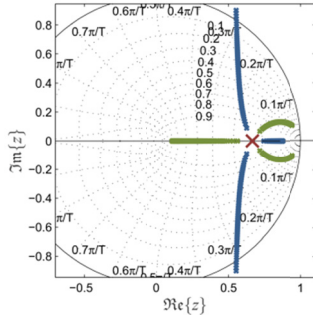


Fig. 5: Variation of Pole Locations

To investigate the small-signal behavior of the closed control loop the variation of the pole locations due to the varying differential inductance with fixed controller parameters is shown in Fig. 5. In steady state operation the inductances in (4) and (5) converge to the differential inductances in (6). If the plant time constant is smaller than the nominal value used to calculate the controller parameters in the range of 0.3 ... 1, the pole locations move away from the specifies position marked with "x" along the blue trajectory. In this case the time constant of the real pole increases and the complex conjugate poles exhibit oscillating behavior with high characteristic frequencies and small decay times. If the plant time constant is higher than the nominal value used to calculate the controller parameters in the range of 1...3, the pole location move away from the specified position along the green trajectory. In this case the time constant of the real pole decreases and the complex conjugate poles exhibit oscillating behavior with small characteristic frequencies and high decay times. Note that the complex conjugate poles become instable in the case 0.3 ... 1 although the absolute inductance variation is considerably smaller than in the case 1... 3.

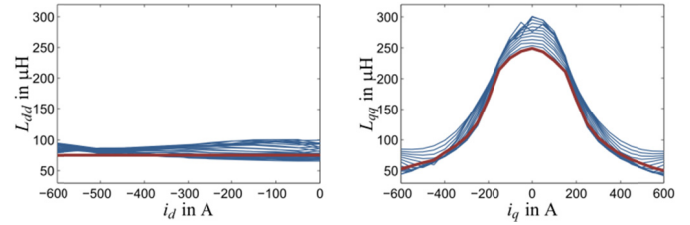


Fig. 6: Inductances and Values used for Gain Adaption

Since the poles cannot be kept exactly at the specified locations a permitted area can be defined so that the variation of the pole locations is tolerable as long as they remain within this area. It can be seen that in both cases shown in Fig. 5 one of the poles remains real while the other poles become complex conjugates. Thus, the permitted area can be defined according to a maximal and minimal time constant of the real pole and decay time of the complex poles and a minimal damping of the complex poles. Although the differential inductances L_{da} and L_{dq} depend on both current components a simplified adaptation strategy can be used with the controller parameters for each current component depending only on that one current component. In Fig. 6 the inductances $L_{da}(i_d, i_q = \text{const})$ and $L_{dq}(i_d = \text{const}, i_q)$ are shown in blue. It can be seen that the inductance L_{da} shows only little variation due to the bigger effective air gap while the inductance L_{dq} strongly depends on the current i_q . Thus a constant value is used to calculate the controller parameters for the current component i_d and a varying value is used to calculate the controller parameters for the current component i_q , both shown as red bold line in Fig. 6. The resulting variation of the pole locations as well as the permitted area is shown in Fig. 7. The permitted area follows from a minimal damping of $d_{min} = 0.5$, a minimal time constant of $T_{min} = 200 \mu\text{s}$ and a maximal time constant of $T_{max} = 8 \text{ ms}$. It can be seen that the poles of both current control loops remain within the permitted area which means a good disturbance rejection in steady-state operation.

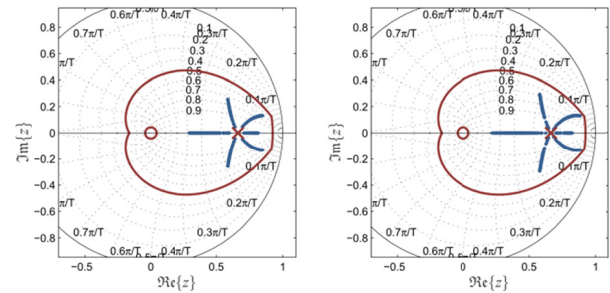


Fig. 7: Pole Locations within the Permitted Area (i_d -controller left, i_q -controller right)

Mostly gain scheduling controllers are utilized to react on conditions that change with significantly slower dynamics than the control loops dynamics. Thus the variation of the controller parameter during transient operations can be neglected and the controller mostly behaves like a controller with constant parameters. However the differential inductances directly depend on the currents and thus are

changing with the same dynamics. Thus the large-signal behavior of the controller has to be investigated.

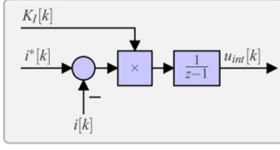


Fig. 8: Integrator Realization 1

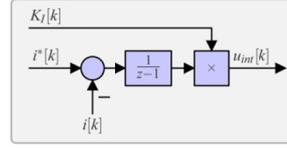


Fig. 9: Integrator Realization 2

In the investigation of the large signal-behavior the way the PI-controllers integrators are realized is crucial. Two possible realizations are shown in Fig. 8 and Fig. 9. In Fig. 8 the measured error at the time instant k is multiplied with the current integrator gain $K_I[k]$ and the resulting product is integrated. In Fig. 9 the measured error is integrated and the sum at the integrator output is multiplied with the current integrator gain $K_I[k]$.

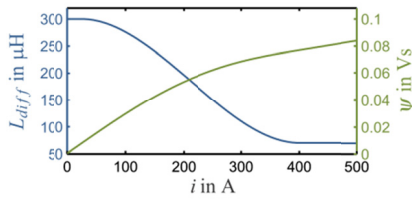


Fig. 10: Magnetization Curve of Simplified Model

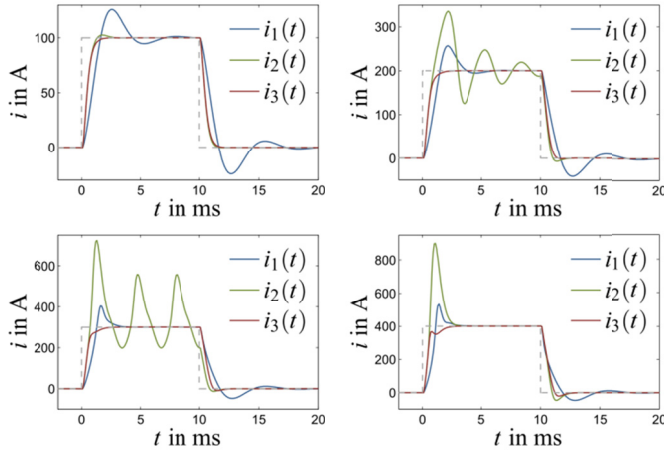


Fig. 11: Step Responses of Simplified Model

To analyze the behavior of both controller realizations in transient operation a simplified model with the magnetization curve shown in Fig. 10 is used. In Fig. 11 some step responses of the simplified model are exemplary shown. The step response of a controller with fixed parameters that is designed to be robust to parameter variations is $i_1(t)$, the step response of the adaptive controller with the integrator realized according Fig. 8 is $i_2(t)$ and the step response of the controller with the integrator realized according to Fig. 9 is $i_3(t)$. Since the pole locations move according to Fig. 5 if a controller with fixes parameters is used, high overshoot occurs. If the current reference moves from an operation point with small saturation to an operation point with high saturation the adaptive controller with the integrator realized

according to Fig. 8 shows a high overshoot and even can get instable. This is because a high measured error occurs immediately after the reference step that is multiplied with a high integrator gain so that a high sum results at the integrator output. When the actual current increases, the inductance and the integrator gain decrease and the high integrator sum has to be reduced with a small measured error and a small integrator gain, which takes time. In contrast the integrator realized according to Fig. 9 provides high dynamic and completely avoids overshoots. Because the measured error is integrated and the integrator output is multiplied with the gain calculated from the current inductance value, the controller goes quickly into steady-state operation. Thus the integrator realization according to Fig. 9 is well suited to adapt the closed loop dynamics to inductance variations due to saturation effects.

EXPERIMENTAL RESULTS

The presented control scheme has been implemented on a dSpace rapid-prototyping system with the current control loop realized according to Fig. 4. The induced voltages are calculated based on lookup tables that have been obtained from measurements thus accurately considering saturation effect in the decoupling of the current components. The dynamical cross-coupling due to cross-saturation effects has been neglected. The PI-controller for the current i_d has fixed controller parameters while the integrator gain of the controller for the current i_q is adapted according to the curve shown in Fig. 6 with the integrator realized according to Fig. 9. The integrator gain is determined using a lookup table from the measured currents. Additionally a controller with fixed parameters for both current components has been realized to exhibit the benefit of the adaptive controller.

To demonstrate the performance of the presented current controller, reference values have been applied manually to the inner current control loop. In Fig. 12 and Fig. 13 reference steps have been applied to the current component i_q while the reference for the current component i_d was kept constant at a speed of 3000 1/min and 12000 1/min. Both measurements show that due to the accurate decoupling almost no disturbance is seen on the current component i_d . Moreover, one can see that the adaptive controller has a higher dynamic than the controller with fixed parameters and avoids overshoot. In Fig. 13 the motor is running with maximum speed so that the controller operates with 10 samplings per electrical fundamental. Since the controller is realized based on the presented discrete-time model that properly describes the effect of a small number of samplings per electrical fundamental, stable operation and an accurate decoupling of the current components are achieved.

In Fig. 14 the disturbance rejection of the presented control scheme is shown. Therefore a voltage step of 25 V is added to the controller output at a speed of 500 1/min. It is obvious that the adaptive controller has a better disturbance rejection than the controller with fixed parameters.

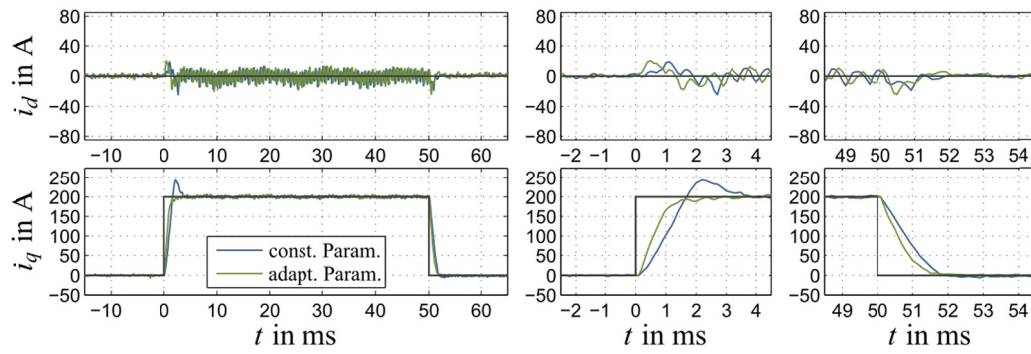


Fig. 12: Measured Step Response at a Speed of 3000 1/min

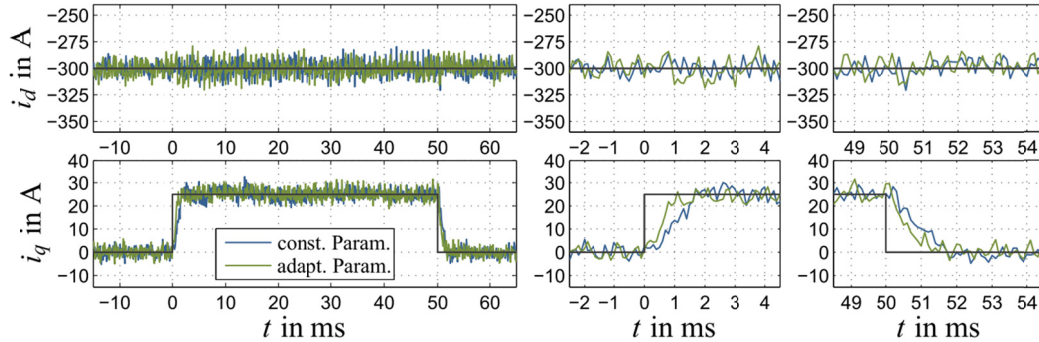


Fig. 13: Measured Step Response with a Speed of 12000 /1min

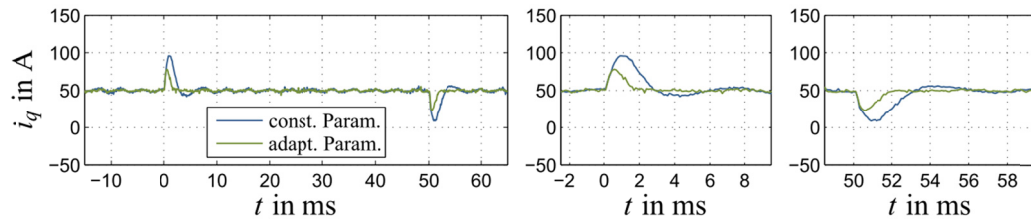


Fig. 14: Measured Disturbance Rejection at a Speed of 500 1/min

CONCLUSION

The presented discrete-time IPMSM model is obtained by discretization of the flux differential equation in stator coordinates and transformation of the resulting flux difference equation into the rotating reference frame. The impact of saturation effects is considered in the calculation of the induced voltages as well as regarding the currents dynamics. The impact of cross-coupling due to cross-saturation effects has been neglected. The derived motor model accurately describes the behavior of the motor in the whole speed range even with a small number of samplings per electrical fundamental at high speeds. The controller parameters are calculated considering the controller dead time that results from regular sampling. The integrator gain of the controller for the current component i_q is adapted to the varying plant time constant. Since the plant time constant cannot be known exactly, the controller is robust to unknown parameter variation due to saturation effects. With the presented integrator realization the current dynamics and the disturbance rejection are improved and overshoot is

completely avoided. Although the dynamics of both current components depend on both current components with the presented adaption of the controller parameters an improvement is achieved with a fixed gain for the d-current controller and an adaptive integrator gain for the q-current controller.

REFERENCES

- [1] Peters W., Huber T., Böcker J., „Control Realization for an Interior Permanent Magnet Synchronous Motor (IPMSM) in Automotive Drive Train”, PCIM 2011.
- [2] M. Meyer and J. Böcker, “Optimum Control for Interior Permanent Magnet Synchronous Motors (IPMSM) in Constant Torque and Flux Weakening Range”, EPE-PEMC 2006.
- [3] D. Pohlenz and J. Böcker, “Efficiency Improvement of an IPMSM using Maximum Efficiency Operating Strategy”, EPE-PEMC 2010.
- [4] T. Windisch, W. Hofmann, “Loss Minimization of an IPMSM Drive Using Precalculated Optimized Current References,” IECON 2011.
- [5] W. Peters, O. Wallscheid, J. Böcker, “A Precise Open Loop Torque Control for an Interior Permanent Magnet Synchronous Motor (IPMSM) considering Iron Losses”, IECON 2012.
- [6] Böcker J., “Discrete-Time Model of an Induction Motor”, ETEP Vol. 1, No. 2, March/April 1991